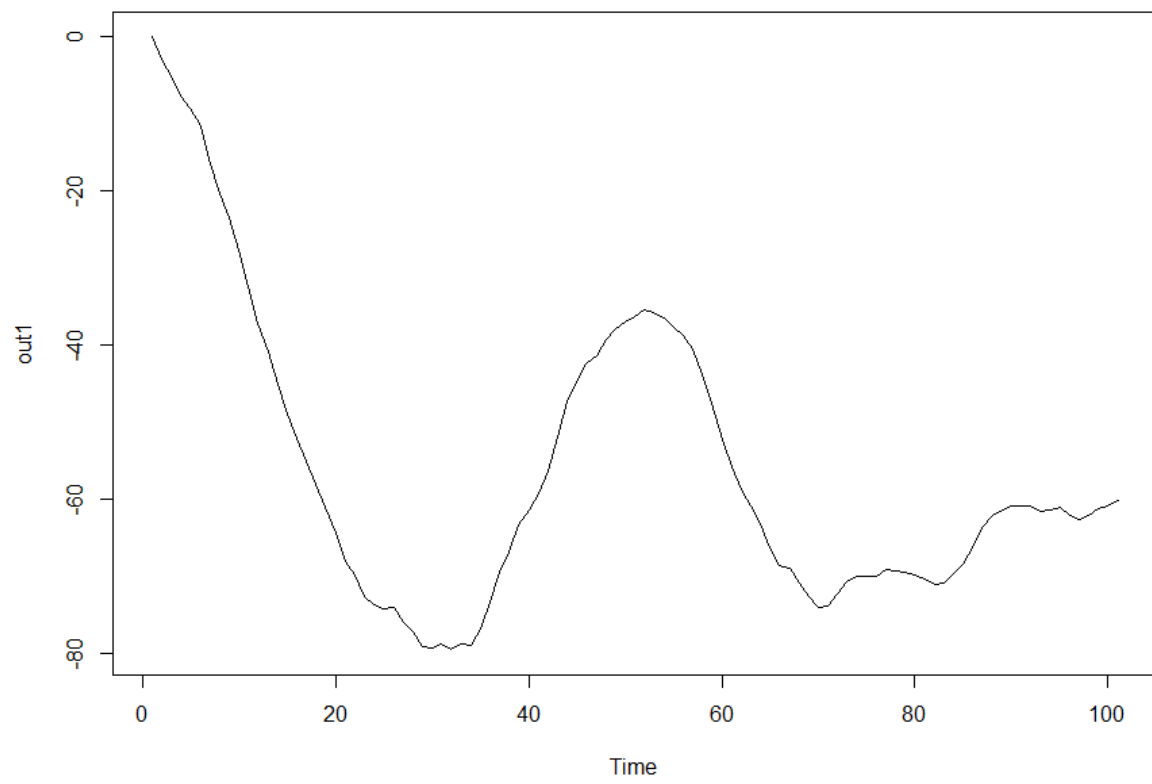
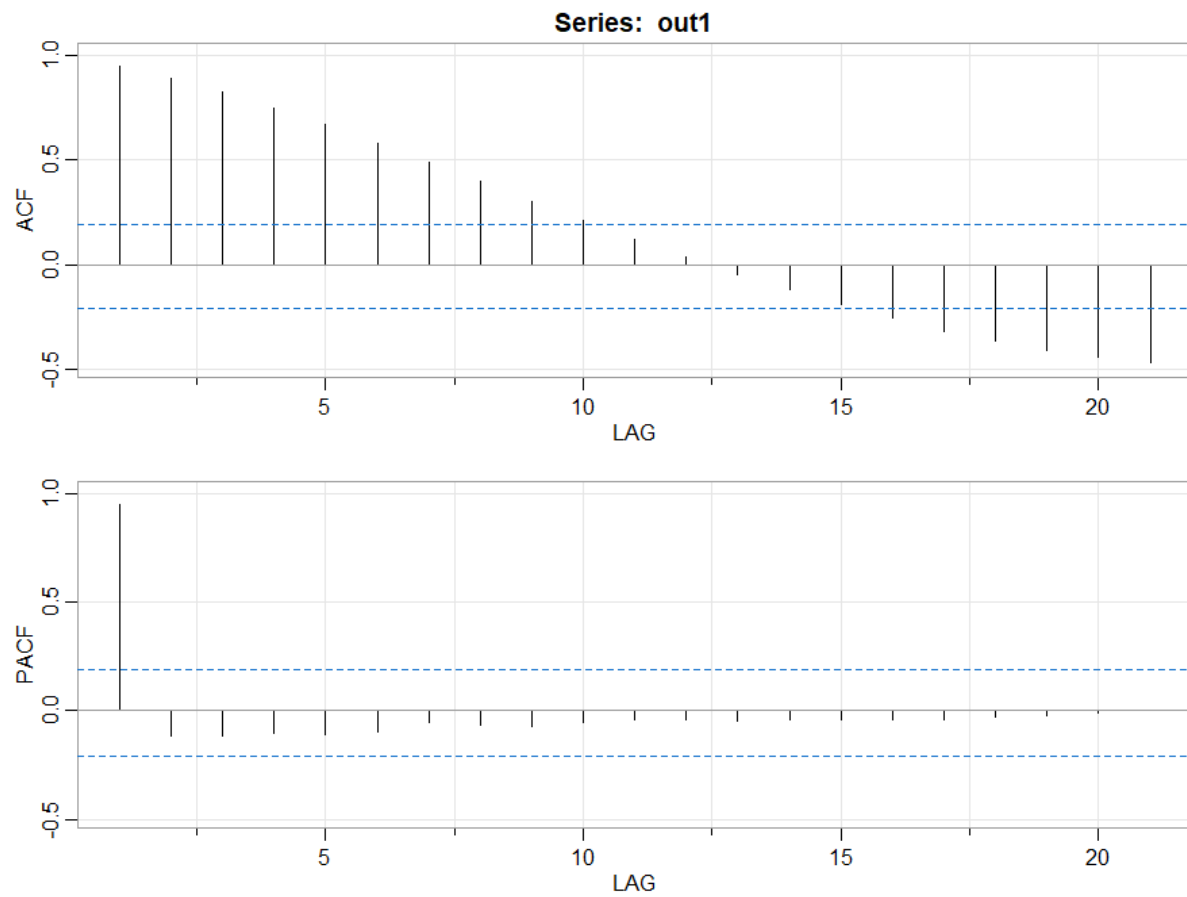


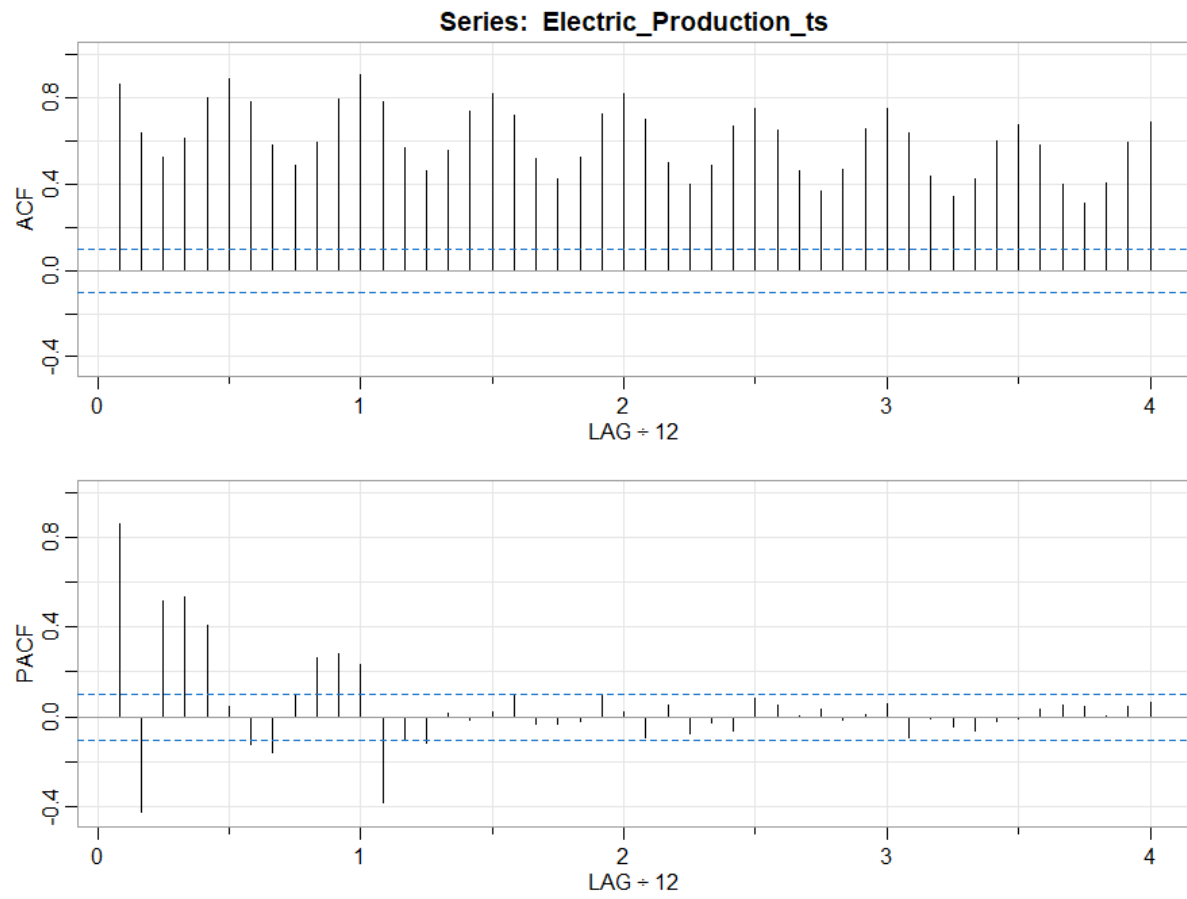
Question 1

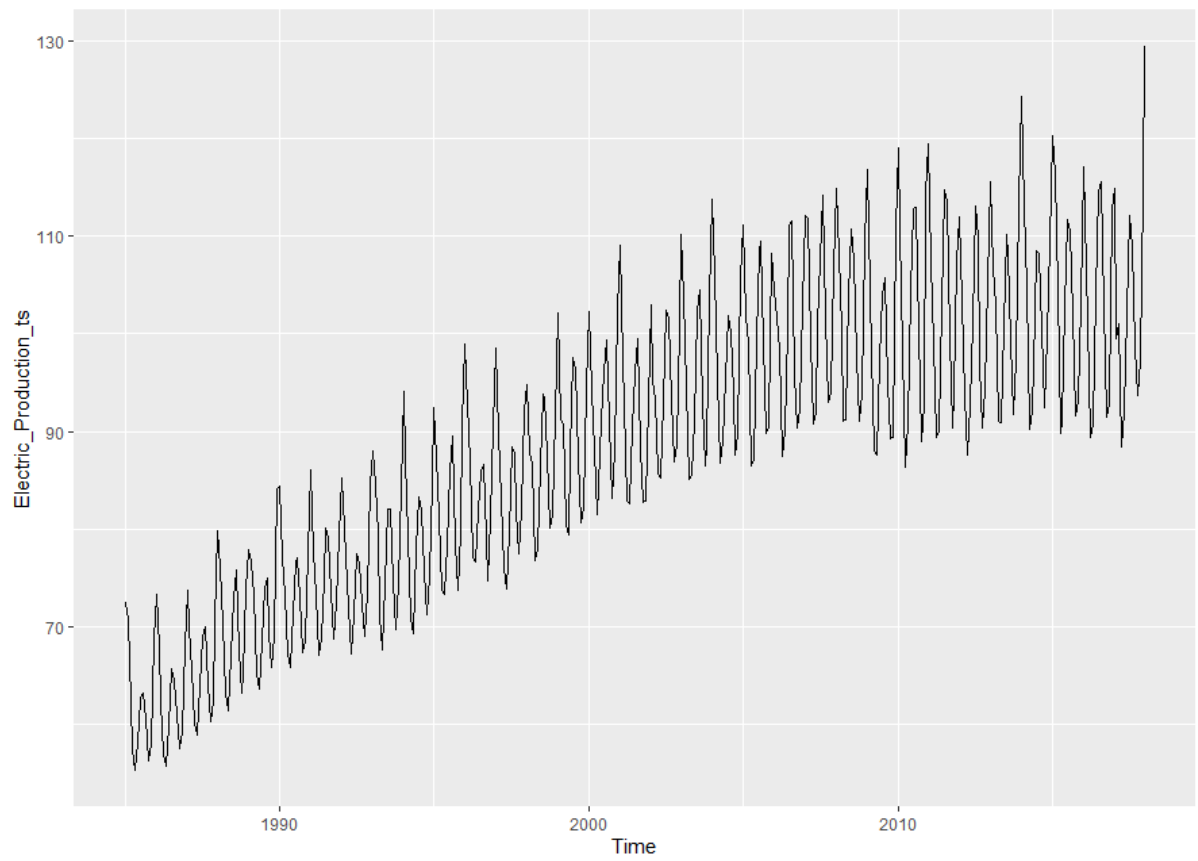




It seems a polynomial function and yes its ARIMA(1,1,0)

Question 2

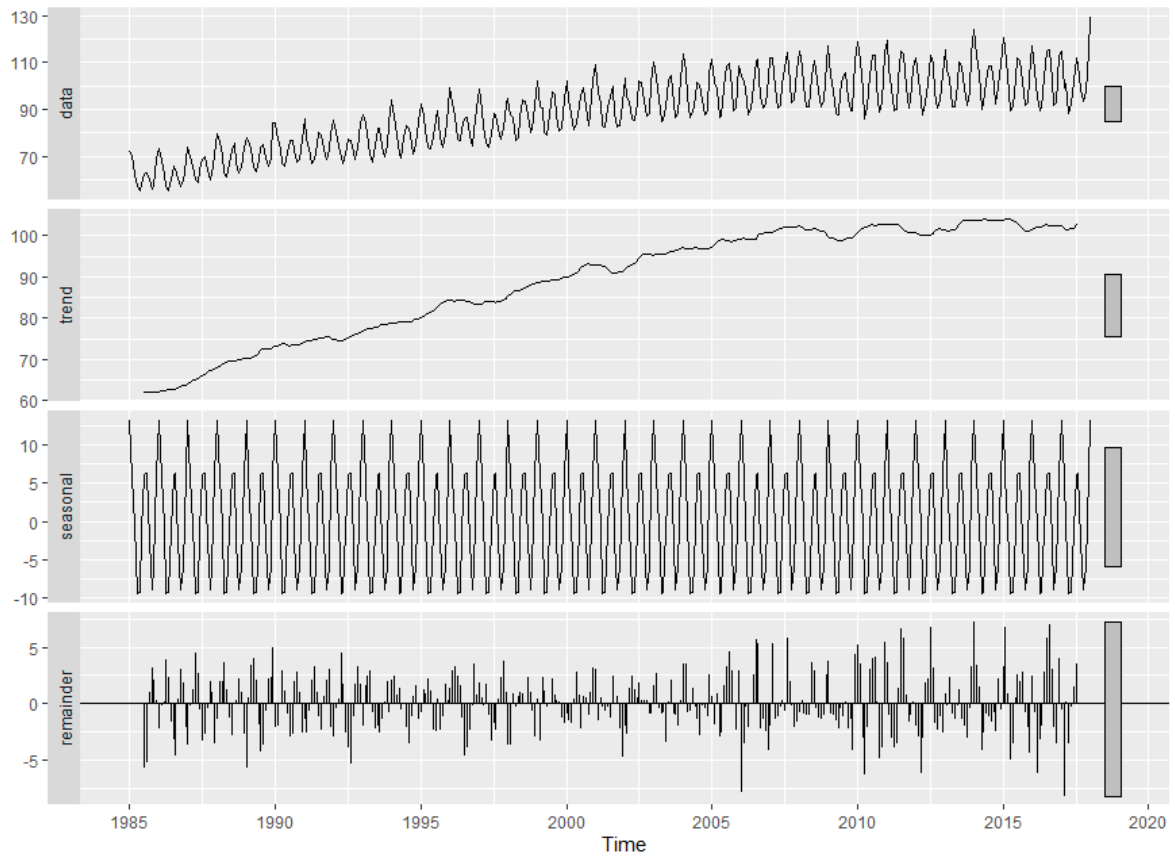




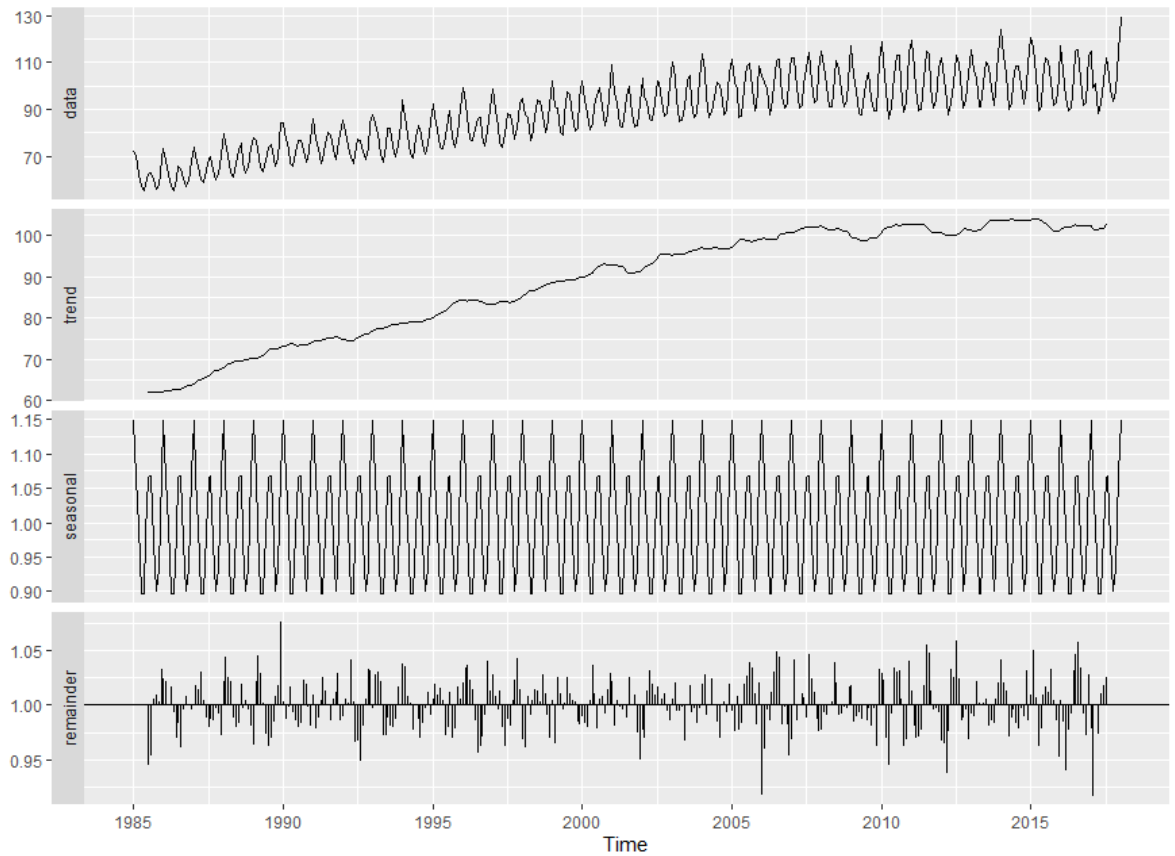
The graph has extended significant lines till end has seasonality and trend too. It seems arima model.

After decomposing

Decomposition of additive time series



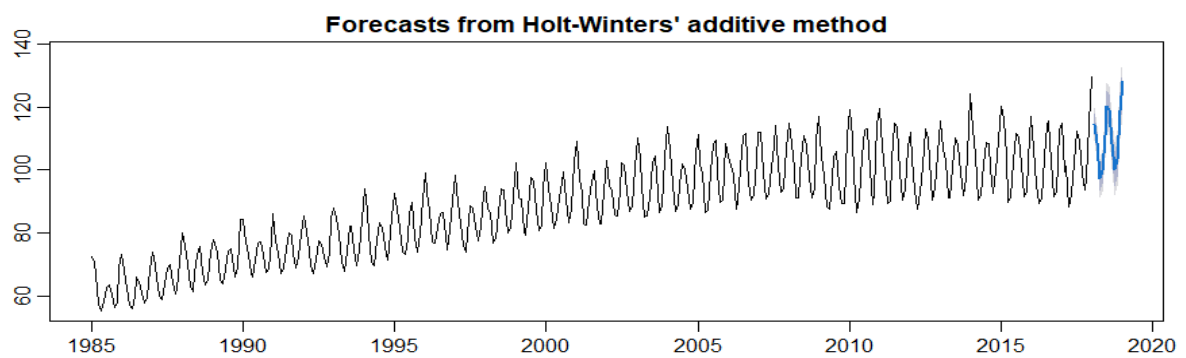
Decomposition of multiplicative time series



Both the models seems same but for simplicity will go with additive as best model.

Periodicity is 12

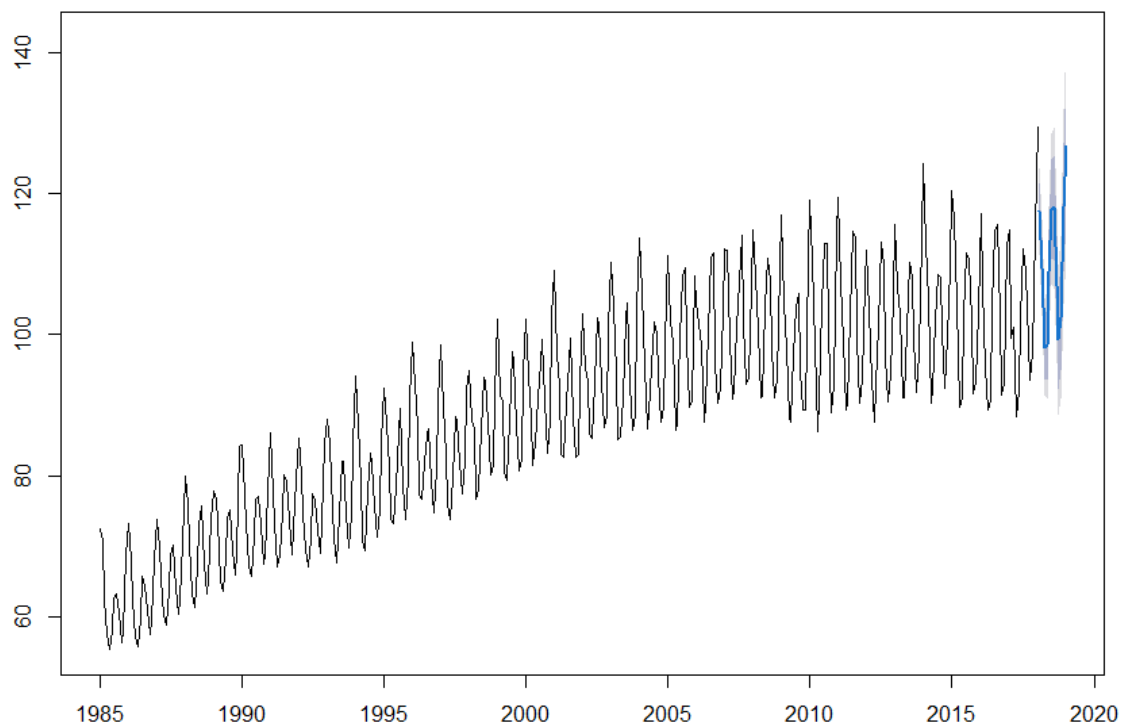
Since the data has seasonal and trend component I have used HW model.



ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
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Training set	0.04583001	2.593804	2.040593	-0.005969128	2.270192	0.7150104	0.1989171
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Forecasts from Damped Holt-Winters' multiplicative method



ME RMSE MAE MPE MAPE MASE ACF1

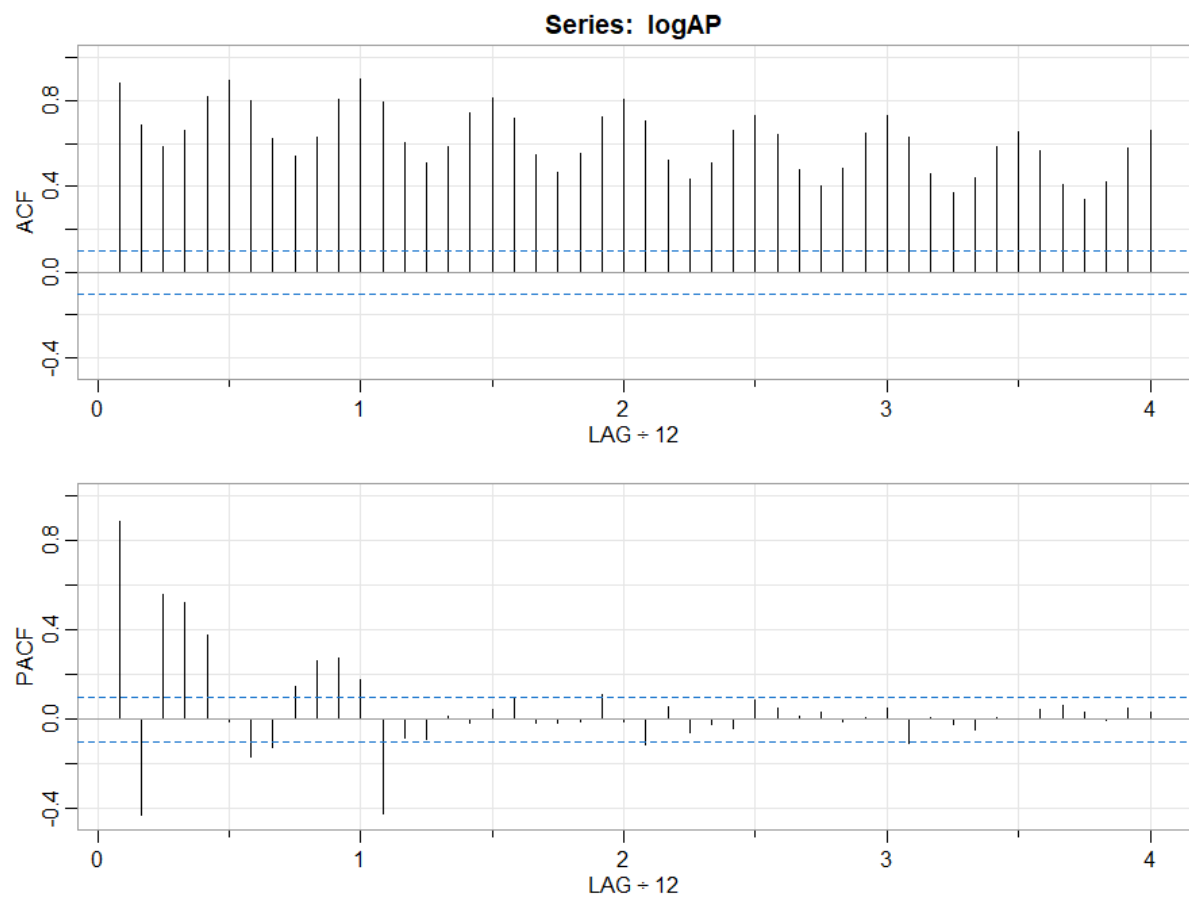
Training set 0.1554166 2.381831 1.831732 0.1330513 2.045588 0.6418266 0.08297627

Multiplicative is better with low MAPE

Data is not stationary

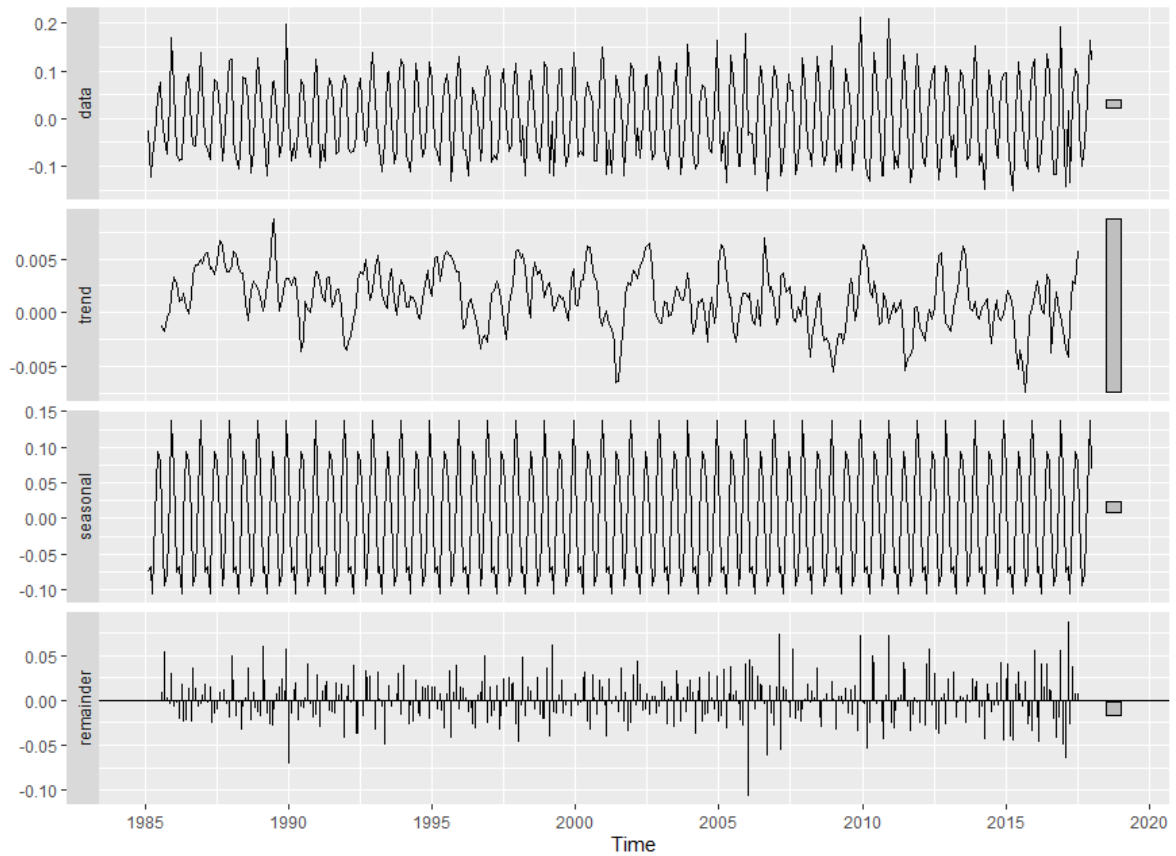
With

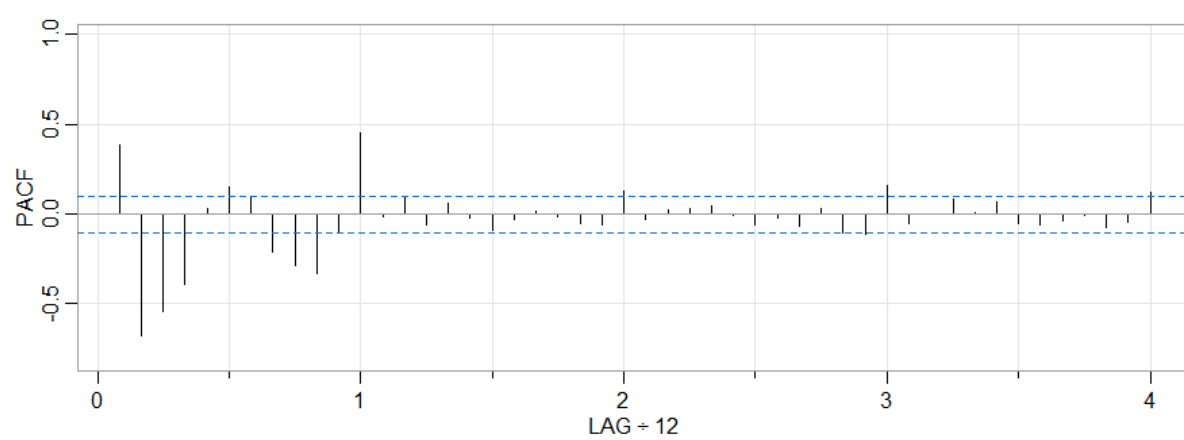
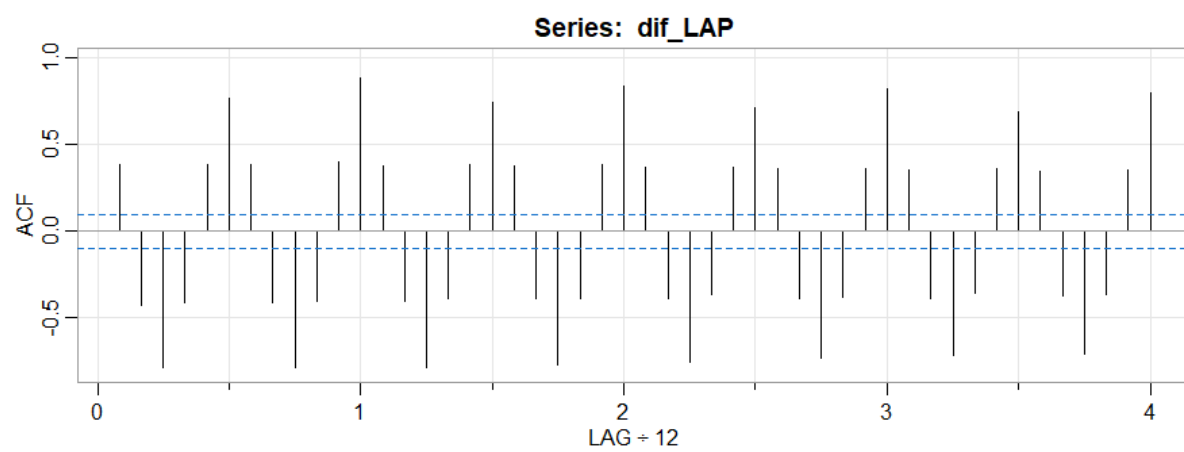
p-value = 0.01

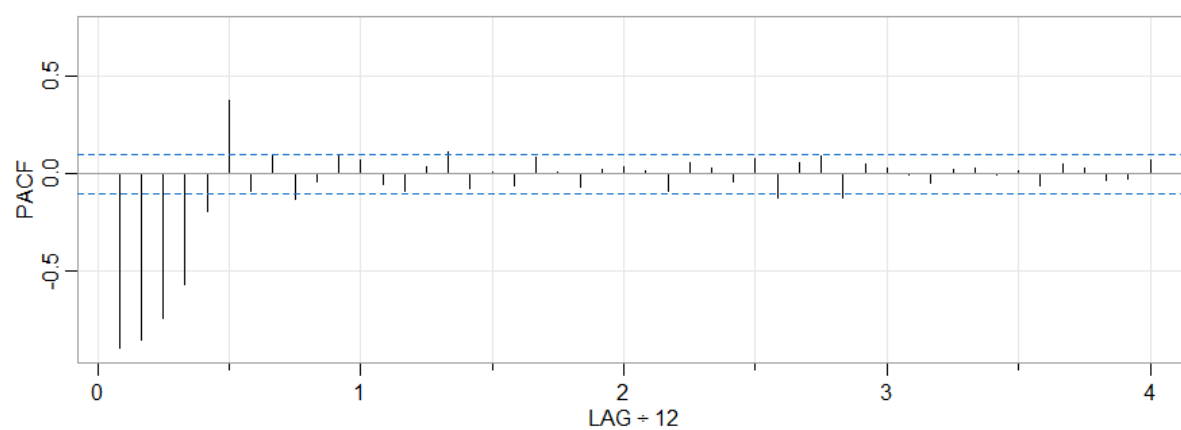
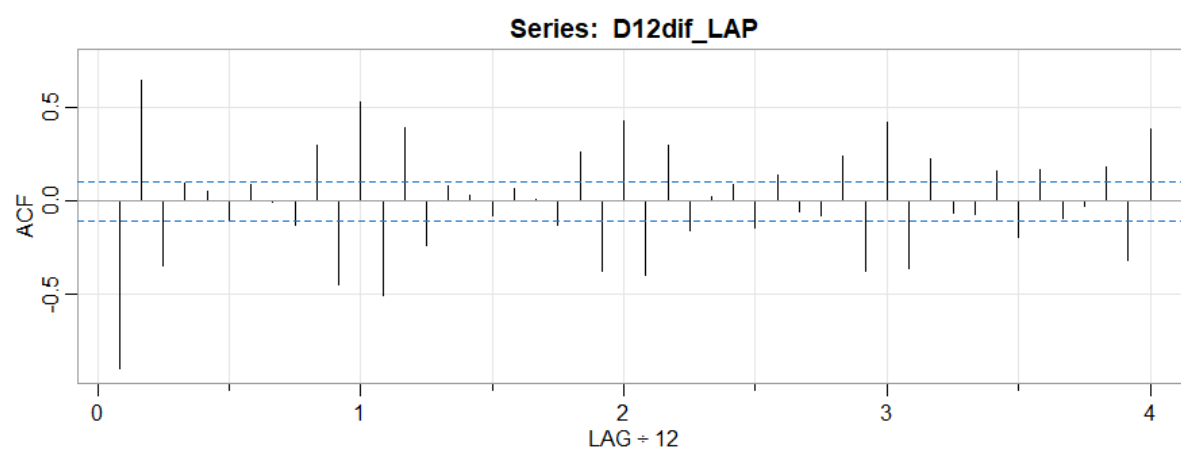


Removing trend and variability.

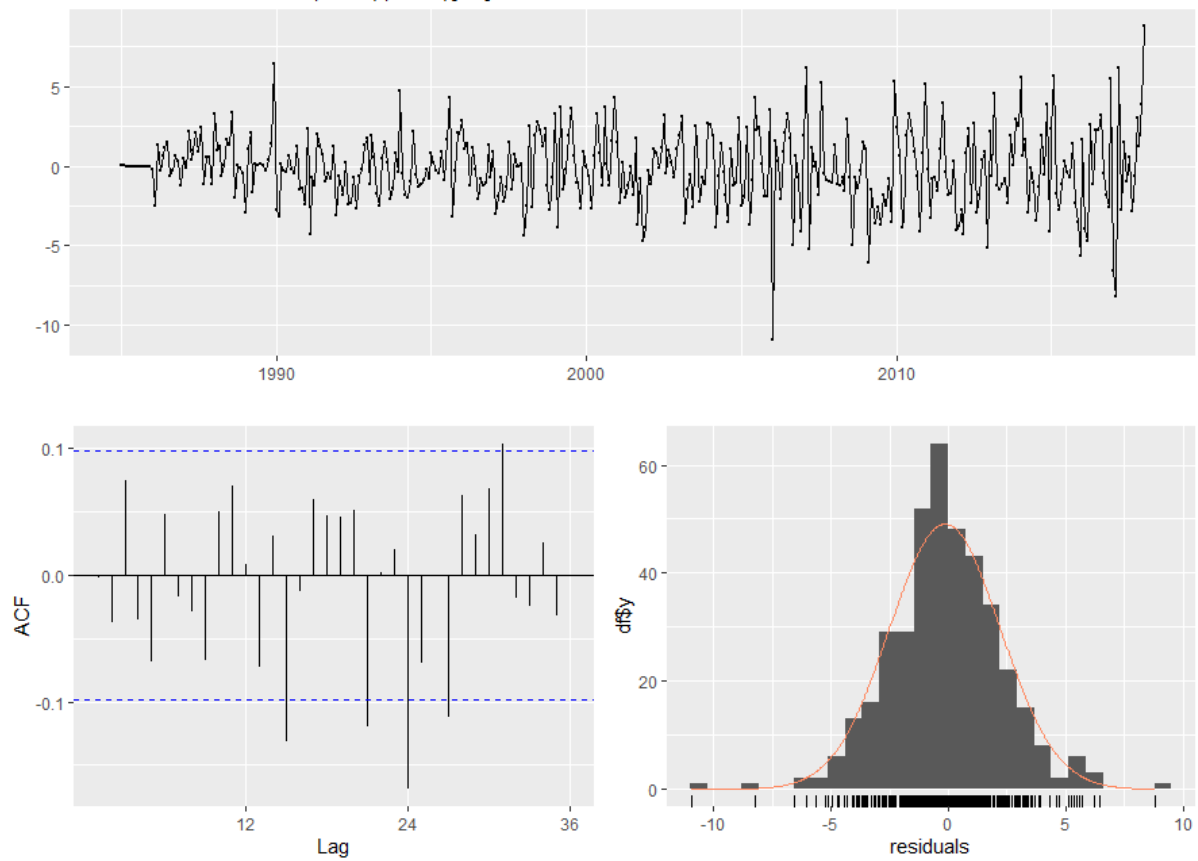
Decomposition of additive time series



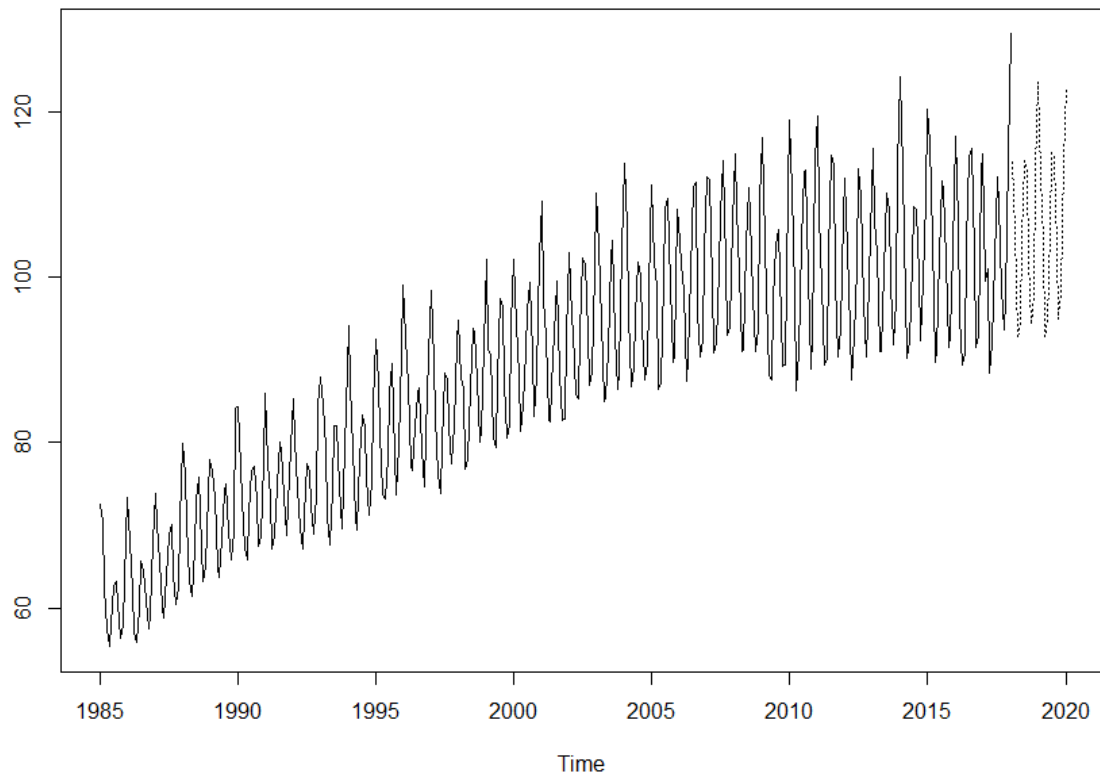




Residuals from ARIMA(2,1,1)(0,1,1)[12]



	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2018	113.87860	105.59452	92.69422	94.23518	104.75129	114.13697	112.90153	102.10169	94.35737	97.66940	113.02747	
2019	123.61176	112.42324	104.10709	92.90637	94.59793	105.63320	115.14190	114.52368	103.35478	94.91081	98.20944	113.86562
2020	122.84458											



Forecasts from ARIMA(1,1,1)(1,1,2)[12]

